Lab 7 - Fuzz Testing Report

Transit Fare Application | Hypothesis and pytest

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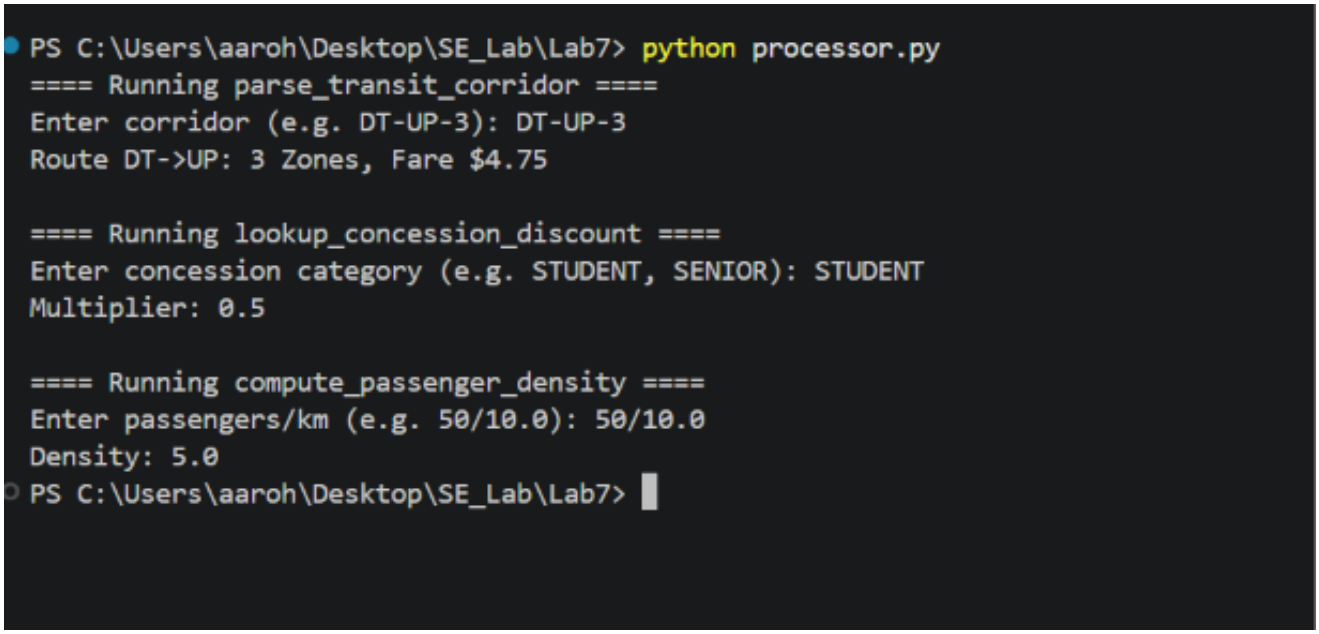
Class: 5A

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This report documents the original transit fare application execution, failures discovered through property-based fuzz testing, the input-validation fixes, and the final test results.

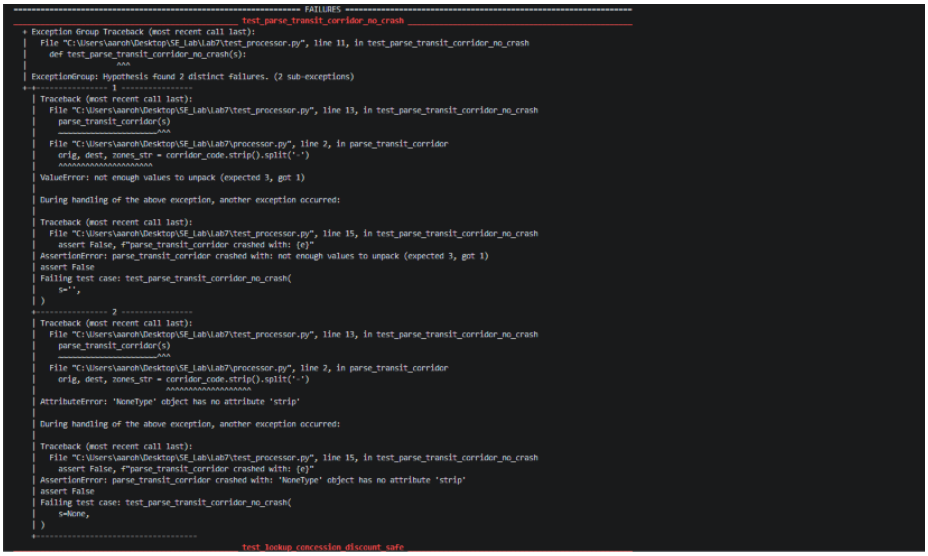
1. Original application execution

The original application was run with valid sample inputs: DT-UP-3 for the transit corridor, STUDENT for the concession category, and 50/10.0 for passenger density.



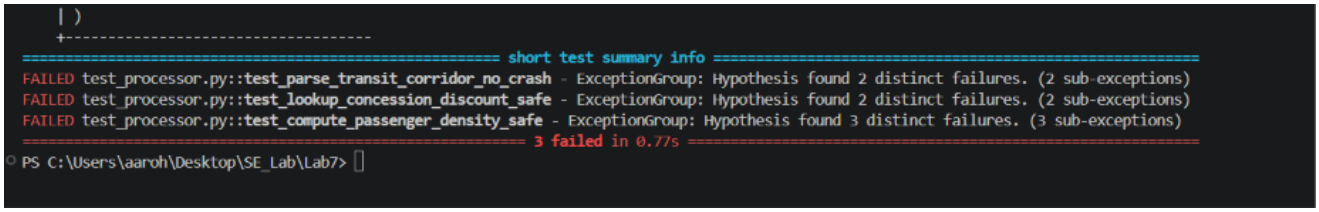
2. Initial Hypothesis failures

The initial fuzz tests exposed failures in all three functions: `parse_transit_corridor`, `lookup_concession_discount`, and `compute_passenger_density`. Hypothesis generated malformed and missing inputs that caused the original implementation to raise exceptions.



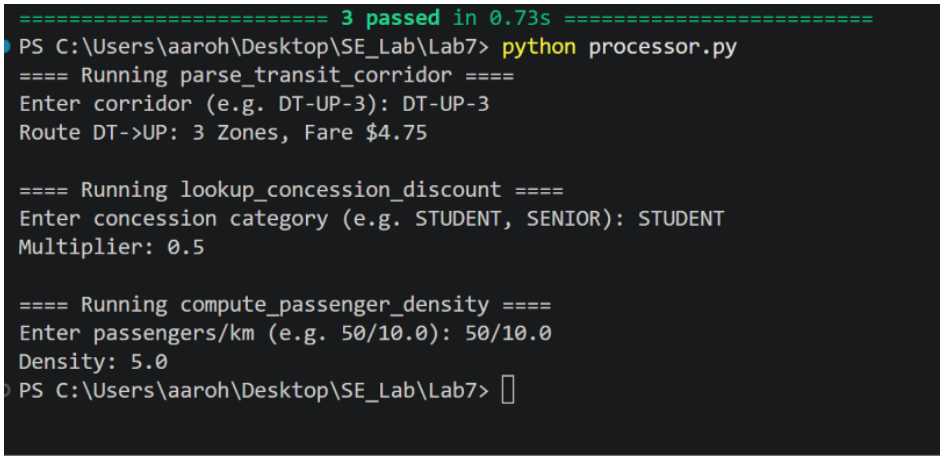
3. Initial failure summary

The pytest summary confirms that all three initial fuzz-test functions failed. The detailed output includes minimized examples such as empty strings, None, and malformed numeric input.



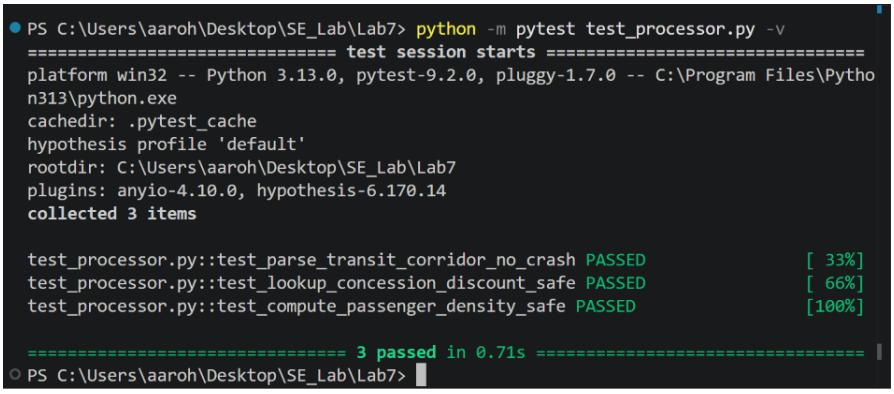
4. Fixed application execution

Input validation was added to all three functions. Invalid corridor codes are handled with an invalid-code message; unsupported concession categories and invalid passenger-density inputs are rejected without uncaught exceptions. The application was then run again with valid sample inputs.



5. Final Hypothesis test results

The updated fuzz-test suite completed successfully: all three tests passed.



Summary of findings

Hypothesis identified malformed corridor codes, unsupported concession categories, missing values, and malformed passenger-to-distance inputs. The application was updated to validate input types and formats and to handle invalid values safely. Final verification: 3 tests passed.